CareerForcePro — Interview Questions Configuration

Backend API Integration Guide (Frontend & QA)

Field	Value
Module	job-service (Spring Boot 3.4.3, Java 17)
Branch	feature/interview-questions-config (base: origin/stage)
Ticket	Configure AI & Video interview questions via multiple input methods
Scope of this doc	Backend endpoints, payloads, responses, validation, QA checklist
Auth	Bearer JWT. Roles: Admin / Super Admin / Hiring Manager / Recruiter (organization)

1. Overview

During Job Opening setup, the AI Questions and Video Questions screens start blank. Users add questions with any of four methods: Generate Default Questions, Upload Excel (.xlsx), Copy-paste multiple questions, or Add one manually. Questions can be edited, deleted, reordered, deleted in bulk, and saved.

Table mapping (important). The two backing tables already existed and were extended rather than replaced:

UI Tab	Backing table	Base path	Shape
AI Questions	job_opening_general_questions	/v1/ai-interview-general-questions	Flat list, ordered by 'sequence', per interview round
Video Questions	job_opening_ai_questions	/v1/ai-interview-questions	Grouped by category, ordered by 'displayOrder', per video round

Note: base paths are relative to the job-service context; the API gateway may prepend a service prefix. Confirm the gateway route in your environment.

2. Data model & question fields

Migration V181 adds a 'source' column to both tables, and a 'display_order' column to job_opening_ai_questions (the AI tab reuses its existing 'sequence' column). Existing rows are backfilled; 'source' defaults to MANUAL.

source (enum) — how the question was added

Value	Meaning
DEFAULT_GENERATED	Produced by Generate Default Questions
COPY_PASTE	Added via the paste text area
MANUAL	Added one-by-one via the Add (+) control
EXCEL_UPLOAD	Imported from an .xlsx upload

Video question object (response)

Field	Type	Notes
id	number	Question id
question	string	Question text (max 100 chars)
categoryId	number	FK to ai_question_categories
categoryName	string	Competency-Based / Open-Ended / Pass/Fail Questions
interviewRoundId	number	Video round id
displayOrder	number	Order within (round + category)
source	enum	See source table above
status	boolean	Active flag
jobOpeningId	number	Job opening id
createdBy	number	User id (nullable)
createdAt	string	yyyy-MM-dd

AI question object (response)

Field	Type	Notes
id	number	Question id
question	string	Question text (max 100 chars)
sequence	number	Order within the round's flat list
source	enum	See source table above
status	boolean	Active flag
interviewRoundId	number	Interview round id
jobOpeningId	number	Job opening id
createdBy	number	User id (nullable)
createdAt	string	ISO-8601 UTC (yyyy-MM-dd'T'HH:mm:ss'Z')

Video categories (seeded in ai_question_categories)

categoryName	Typical id
Competency-Based Questions	3
Open-Ended Questions	2
Pass/Fail Questions	1

Do not hard-code ids — resolve categoryName -> id from the categories dropdown for the org/environment. The ids above reflect the seed insertion order only.

3. Response envelopes

Three shapes are used across the endpoints:

MessageResponse<T> — reads and bulk operations that return data:

```
{ "success": true, "message": "Questions have been added successfully.", "data": { ... } }
```

CustomMessage — simple mutations (generate / reorder / delete-all / single add):

```
{ "message": "Default questions have been generated successfully.", "success": true }
```

Message — error body from the global handler:

```
{ "status": "BAD_REQUEST", "message": "Question must not exceed 100 characters", "code":
"BAD_REQUEST" }
```

4. Shared behaviour

Copy-paste parsing

- Split the text on new lines; trim each line; skip blank lines; preserve order.
- Each resulting question is validated (non-blank, <= 100 chars) and saved with source = COPY_PASTE.
- New items are appended after the current max order in scope.

Generate Default Questions — mode

- **mode=REPLACE** (default): clears existing questions in scope, then generates.
- **mode=ADD** : appends generated questions after existing ones.
- **Cancel** : no request is sent (client-side).

Generated questions have source = DEFAULT_GENERATED. Video generation splits into the three categories; if the backend cannot determine a category it falls back to Open-Ended Questions. Generation calls the existing AI pipeline, so responses may take a few seconds.

Excel upload (.xlsx only)

Tab	Required column	Optional columns
AI Questions	Question Text	Order
Video Questions	Question Text	Category, Order

- Headers are matched case-insensitively. Blank rows are skipped and counted in 'skippedRows'.
- Rows with a question longer than 100 chars are skipped and reported in 'warnings' (the rest still import).
- If an 'Order' value is present it is used as the order; otherwise items are appended sequentially.

Video Excel — category mapping

Excel value (case/spacing-insensitive)	System category
Competency / Competency Based	Competency-Based Questions
Open Ended / Open-Ended	Open-Ended Questions
Pass Fail / Pass/Fail	Pass/Fail Questions
missing or unrecognised	Open-Ended Questions (+ row warning)

Excel upload response — QuestionExcelUploadResponseDto (inside data)

```
{
  "uploadedFileName": "questions.xlsx",
  "importedCount": 8,
  "skippedRows": 2,
  "warnings": [ "Row 4: Category missing or invalid. Question added under Open-Ended
Questions." ],
  "questions": [ { /* saved question object */ } ]
}
```

Ordering & reorder

- Video order is scoped per (jobOpening + round + category). AI order is scoped per (jobOpening + round).
- Reorder sends the full list of {id, order} pairs for one scope; the backend writes the new order.

5. AI Questions tab endpoints

Base path: /v1/ai-interview-general-questions

Method	Path	Purpose
GET	/getAllByJobId	Fetch all AI questions for a job (sorted by sequence)
POST	/save	Add one question manually (MANUAL)
PUT	/edit	Edit a question
DELETE	/delete	Delete one question
POST	/paste	Add pasted questions (COPY_PASTE) [NEW]
POST	/upload-excel	Import from .xlsx (EXCEL_UPLOAD) [NEW]
POST	/generate-questions	Generate defaults, mode=ADD|REPLACE [NEW]
PUT	/reorder	Persist new order [NEW]
DELETE	/delete-all	Delete all questions of a round [NEW]

POST /paste [NEW]

Query: jobOpeningId, interviewRoundId. Body:

```
{ "text": "First question\nSecond question\nThird question" }
```

Response (201): MessageResponse<list of AI question objects> with the saved questions.

POST /upload-excel [NEW]

multipart/form-data. Parts: file (.xlsx), jobOpeningId, interviewRoundId.

```
curl -F "file=@ai_questions.xlsx" \
     -F "jobOpeningId=101" -F "interviewRoundId=55" \
     .../v1/ai-interview-general-questions/upload-excel
```

Response (200): MessageResponse<QuestionExcelUploadResponseDto>.

POST /generate-questions [NEW]

Query: jobOpeningId, interviewRoundId, mode (ADD|REPLACE, default REPLACE). Response (200): CustomMessage.

PUT /reorder [NEW]

Query: jobOpeningId, interviewRoundId. Body:

```
{ "items": [ { "id": 12, "order": 1 }, { "id": 11, "order": 2 } ] }
```

Response (200): CustomMessage.

DELETE /delete-all [NEW]

Query: jobOpeningId, interviewRoundId. Response (200): CustomMessage.

Existing single-add / edit

POST /save & PUT /edit body (JobOpeningAIGeneralQuestionDto):

```
{ "question": "Explain dependency injection", "sequence": 3 }
```

6. Video Questions tab endpoints

Base path: /v1/ai-interview-questions

Method	Path	Purpose
GET	/getAllByRoundID	Fetch questions for a round, grouped by category
POST	/save	Add one question manually (MANUAL)
PUT	/edit	Edit a question
DELETE	/delete	Delete one question
POST	/paste	Add pasted questions to a category (COPY_PASTE) [NEW]
POST	/upload-excel	Import from .xlsx, mapped to categories [NEW]
POST	/generate-questions	Generate defaults, mode=ADD|REPLACE [NEW]
PUT	/reorder	Persist new order within a category [NEW]
DELETE	/delete-all	Delete all questions in one category [NEW]

GET /getAllByRoundID

Query: interviewRoundId, jobOpeningId. Returns a category-keyed map (CategorizedQuestionsDTO):

```
{
  "categorizedQuestions": {
    "Competency-Based Questions": [ { "id": 1, "question": "...", "categoryId": 3,
"displayOrder": 1, "source": "DEFAULT_GENERATED", ... } ],
    "Pass/Fail Questions": [ ... ],
    "Open-Ended Questions": [ ... ]
  }
}
```

POST /paste [NEW]

Query: jobOpeningId, interviewRoundId, categoryId. Body:

```
{ "text": "Describe a conflict you resolved\nHow do you prioritise work?" }
```

Response (201): MessageResponse<list of video question objects>. source = COPY_PASTE; all rows go under the given categoryId.

POST /upload-excel [NEW]

multipart/form-data. Parts: file (.xlsx), jobOpeningId, interviewRoundId. Category comes from the file (see mapping). Response (200): MessageResponse<QuestionExcelUploadResponseDto>.

POST /generate-questions [NEW]

Query: jobOpeningId, interviewRoundId, aiJobDescriptionId, mode (ADD|REPLACE, default REPLACE). Response (200): CustomMessage.

aiJobDescriptionId is required — it identifies the job description the AI uses to generate questions.

PUT /reorder [NEW]

Query: jobOpeningId, interviewRoundId, categoryId. Body: same {items:[{id,order}]} shape. Reorder is per category.

DELETE /delete-all [NEW]

Query: jobOpeningId, interviewRoundId, categoryId. Deletes only that category for that round (other categories / rounds are untouched). Response (200): CustomMessage.

Existing single-add / edit

Note the capitalised query param **`JobOpeningId`** on /save.

POST /save?JobOpeningId=&interviewRoundId= & PUT /edit?questionID=&interviewRoundId= body (JobOpeningAIInterviewQuestionDTO):

```
{ "categoryId": 3, "question": "Rate the candidate on teamwork" }
```

7. Validation & error responses

All errors return the Message shape { status, message, code } with an appropriate HTTP status.

Scenario	HTTP	code	Message (example)
Blank question (manual/paste body)	400	VALIDATION_ERROR	Question must not be blank
Question > 100 chars (manual/edit/paste)	400	BAD_REQUEST	Question must not exceed 100 characters
Excel missing 'Question Text' column	400	BAD_REQUEST	Required column 'Question Text' is missing.
Non-.xlsx / corrupt file	400	VALIDATION_ERROR	INVALID_FILE: Unsupported file format. Please upload a valid .xlsx file.
Missing required query param	400	MISSING_PARAMETER	Missing required parameter: interviewRoundId
Invalid mode value	400	VALIDATION_ERROR	Invalid value 'FOO' ... Allowed values are: [ADD, REPLACE]
Job / round / question not found	404	RESOURCE_NOT_FOUND	Job opening not found!
Reorder id not in scope	404	RESOURCE_NOT_FOUND	Question not found in this category: 99
Not authenticated / wrong role	401 / 403	-	Access denied

Excel behaviour that is NOT an error: blank rows are skipped (counted in skippedRows); missing/invalid category defaults to Open-Ended with a warning; over-length rows are skipped with a warning. These come back in the 200 response, not as errors.

8. QA checklist (Acceptance Criteria)

Backend-verifiable behaviour for each AC. AC1/AC2/AC13 are UI-state criteria verified at the API-contract level.

AC	What to verify	Expected
AC3	POST AI /paste with multi-line text incl. blank lines	Each non-blank line saved in order; source=COPY_PASTE; input order preserved
AC4	POST Video /paste under a category	Lines saved under that category & round; source=COPY_PASTE
AC5	POST /save one question	One question saved; source=MANUAL; editable in fetch
AC6	POST AI /upload-excel valid file with a blank row	Valid rows imported; skippedRows counts blanks; uploadedFileName echoed
AC7	POST Video /upload-excel with mixed/blank categories	Mapped to correct categories; invalid -> Open-Ended + warning; linked to round
AC8	POST /generate-questions (ADD & REPLACE)	REPLACE clears then generates; ADD appends; source=DEFAULT_GENERATED
AC9	PUT /edit question text	Updated text persists on fetch; length validated
AC10	DELETE /delete	Question removed on fetch

AC	What to verify	Expected
AC11	DELETE /delete-all	Video: only that category cleared (other categories/rounds intact); AI: round cleared
AC12	PUT /reorder	New order persists on fetch; scoped per category (video)
AC13	Any mutation then fetch	Changes persisted; success envelope returned
AC14	POST /save or /paste with blank	400; nothing saved
AC15	POST /upload-excel with .csv/.txt or corrupt	400 INVALID_FILE; nothing saved

9. Assumptions & out of scope

Assumptions (flag if any differ from FE expectations)

- Max question length is 100 characters (applies to new input; legacy rows may be longer).
- AI-tab questions keep their interview-round association (roundId is not null).
- Generate-when-existing uses mode=ADD|REPLACE; Cancel is client-side (no call).
- Upload endpoints persist immediately (per-action) and return the saved rows; there is no separate 'preview then save' round-trip on the backend.
- Excel 'Order' value, when present, is used verbatim as the order; otherwise items are appended.

Out of scope (not implemented)

- PDF/DOCX parsing; AI auto-categorization of pasted questions; shared/global question library.
- Copying question sets across job openings; bulk import across openings; advanced duplicate-resolution.
- Scoring-criteria auto-generation; candidate-facing preview; all frontend work.